1) a)

$\frac{1}{3}k(k+1)(k+2)+(k+1)(k+2)=(k+1)(k+2)\left(\frac{1}{3}k+1\right)$

$=\frac{1}{3}(k+1)(k+2)(k+3)$

1) b)

$\frac{k}{k+1}+\frac{1}{(k+1)(k+2)}=\frac{k(k+2)}{(k+1)(k+2)}+\frac{1}{(k+1)(k+2)}=\frac{k(k+2)+1}{(k+1)(k+2)}=\frac{k^2+2k+1}{(k+1)(k+2)}=\frac{(k+1)^2}{(k+1)(k+2)}$

$=\frac{k+1}{k+2}$ as req.

1) c)

$(k+1)!-1+(k+1)(k+1)!=(k+1)![1+(k+1)]-1$

$=(k+1)!\,(k+2)-1$

$=(k+2)!-1$ as req.

2) a)

Prove true for $n=1$:

$$\sum_{r=1}^{1}2^{r-1}=2^{1-1}=1$$

$2^1-1=1$

so true for $n=1$

Assume true for $n=k$:

$$\sum_{r=1}^{k}2^{k-1}=2^k-1$$

Prove true for $n=k+1$:

$$\sum_{r=1}^{k+1}2^{k-1}=\sum_{r=1}^{k}2^{k-1}+2^{k+1-1}$$

$=2^k-1+2^k$

$=2\times 2^k-1$

$=2^{k+1}-1$

which is 2^n-1 with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n\geq 1$

2) b)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} 4^r = 4^1 = 4$$

$$\frac{4}{3}(4^1 - 1) = \frac{4}{3} \times 3 = 1$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} 4^r = \frac{4}{3}(4^k - 1)$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} 4^r = \sum_{r=1}^{k} 4^r + 4^{k+1}$$

$$= \frac{4}{3}(4^k - 1) + 4^{k+1}$$

$$= \frac{4}{3}(4^k - 1) + 4 \times 4^k$$

$$= \tfrac{4}{3}(4^k - 1 + 3 \times 4^k)$$

$$= \frac{4}{3}(4 \times 4^k - 1)$$

$$= \frac{4}{3}(4^{k+1} - 1)$$

which is $\frac{4}{3}(4^n - 1)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

3) a)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} r = 1 = 1$$

$$\frac{1}{2} \times 1(1+1) = \frac{1}{2} \times 2 = 1$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} r = \frac{1}{2}k(k+1)$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} r = \sum_{r=1}^{k} r + (k+1)$$

$$= \frac{1}{2}k(k+1) + (k+1)$$

$$= \frac{1}{2}(k+1)[k+2]$$

$$= \frac{1}{2}(k+1)(k+2)$$

which is $\frac{1}{2}n(n+1)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

3) b)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} r^2 = 1^2 = 1$$

$$\frac{1}{6} \times 1(1+1)(2+1) = \frac{1}{6} \times 2 \times 3 = 1$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} r^2 = \frac{1}{6}k(k+1)(2k+1)$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} r^2 = \sum_{r=1}^{k} r^2 + (k+1)^2$$

$$= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2$$

$$= \frac{1}{6}(k+1)[k(2k+1) + 6(k+1)]$$

$$= \frac{1}{6}(k+1)[2k^2 + k + 6k + 6]$$

$$= \frac{1}{6}(k+1)[2k^2 + 7k + 6]$$

$$= \frac{1}{6}(k+1)(k+2)(2k+3)$$

$$= \frac{1}{6}(k+1)(k+1+1)(2(k+1)+1)$$

which is $\frac{1}{6}n(n+1)(2n+1)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

3) c)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} r^3 = 1^3 = 1$$

$$\frac{1}{4} \times 1^2(1+1)^2 = \frac{1}{4} \times 4 = 1$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} r^3 = \sum_{r=1}^{k} r^3 + (k+1)^3$$

$$= \frac{1}{4}k^2(k+1)^2 + (k+1)^3$$

$$= \frac{1}{4}(k+1)^2[k^2 + 4(k+1)]$$

$$= \frac{1}{4}(k+1)^2[k^2 + 4k + 4]$$

$$= \frac{1}{6}(k+1)^2(k+2)^2$$

which is $\frac{1}{4}n^2(n+1)^2$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

4)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} r(r-1) = 1 \times 0 = 0$$

$$\frac{1}{3} \times 1(1-1)(1+1) = 0$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} r(r-1) = \frac{1}{3}k(k-1)(k+1)$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} r(r-1) = \sum_{r=1}^{k} r(r-1) + (k+1)(k+1-1)$$

$$= \frac{1}{3}k(k-1)(k+1) + k(k+1)$$

$$= \frac{1}{3}k(k+1)[(k-1)+3]$$

$$= \frac{1}{3}(k+1)k(k+2)$$

which is $\frac{1}{3}n(n-1)(n+1)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

5)

$-3-4-3+\cdots+n(n-4)$

$=\sum_{r=1}^{n} r(r-4)$

Prove true for $n=1$:

$\sum_{r=1}^{1} r(r-4)=1\times-3=-3$

$\frac{1}{6}\times1\times2\times(2-11)=\frac{1}{6}\times2\times-9=-3$

so true for $n=1$

Assume true for $n=k$:

$\sum_{r=1}^{k} r(r-4)=\frac{1}{6}k(k+1)(2k-11)$

Prove true for $n=k+1$:

$\sum_{r=1}^{k+1} r(r-4)=\sum_{r=1}^{k} r(r-4)+(k+1)(k+1-4)$

$=\frac{1}{6}k(k+1)(2k-11)+(k+1)(k-3)$

$=\frac{1}{6}(k+1)[k(2k-11)+6(k-3)]$

$=\frac{1}{6}(k+1)(2k^2-11k+6k-18)$

$=\frac{1}{3}(k+1)(k+2)(2k-9)$

$=\frac{1}{6}(k+1)(k+2)(2(k+1)-11)$

which is $\frac{1}{6}n(n+1)(2n-11)$ with $k+1$ substituted in.

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n\geq1$

6)

He wrote $k+1$ rather than the $(k+1)^{\text{th}}$ term. So

$\sum_{r=1}^{k+1} r(r+3)=\sum_{r=1}^{k} r(r+3)+(k+1)(k+4)$

7)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{1}{(r+1)(r+2)} = \frac{1}{2 \times 3} = \frac{1}{6}$$

$$\frac{1}{2+4} = \frac{1}{6}$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} \frac{1}{(r+1)(r+2)} = \frac{k}{2k+4}$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{1}{(r+1)(r+2)} = \sum_{r=1}^{k} \frac{1}{(r+1)(r+2)} + \frac{1}{(k+1+1)(k+1+2)}$$

$$= \frac{k}{2k+4} + \frac{1}{(k+2)(k+3)}$$

$$= \frac{k(k+3)}{2(k+2)(k+3)} + \frac{2}{2(k+2)(k+3)}$$

$$= \frac{k^2+3k+2}{2(k+2)(k+3)}$$

$$= \frac{(k+2)(k+1)}{2(k+2)(k+3)}$$

$$= \frac{k+1}{2(k+3)}$$

$$= \frac{k+1}{2k+6}$$

$$= \frac{k+1}{2(k+1)+4}$$

which is $\frac{n}{2n+4}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

8)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{1}{(r+2)(r+4)} = \frac{1}{3 \times 5} = \frac{1}{15}$$

$$\frac{1(7+25)}{24 \times 4 \times 5} = \frac{32}{480} = \frac{1}{15}$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} \frac{1}{(r+2)(r+4)} = \frac{k(7k+25)}{24(k+3)(k+4)}$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{1}{(r+2)(r+4)} = \sum_{r=1}^{k} \frac{1}{(r+2)(r+4)} + \frac{1}{(k+1+2)(k+1+4)}$$

$$= \frac{k(7k+25)}{24(k+3)(k+4)} + \frac{1}{(k+3)(k+5)}$$

$$= \frac{k(7k+25)(k+5)}{24(k+3)(k+4)(k+5)} + \frac{24(k+4)}{24(k+3)(k+4)(k+5)}$$

$$= \frac{k(7k^2+60k+125)+24(k+4)}{24(k+3)(k+4)(k+5)}$$

$$= \frac{7k^3+60k^2+125k+24k+96}{24(k+3)(k+4)(k+5)}$$

$$= \frac{7k^3+60k^2+149k+96}{24(k+3)(k+4)(k+5)}$$

$$= \frac{(k+3)(7k^2+39k+32)}{24(k+3)(k+4)(k+5)}$$

$$= \frac{7k^2+39k+32}{24(k+4)(k+5)}$$

$$= \frac{(k+1)(7k+32)}{24(k+4)(k+5)}$$

$$= \frac{(k+1)(7(k+1)+25)}{24(k+4)(k+5)}$$

which is $\frac{n(7n+25)}{24(n+3)(n+4)}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

9) a)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{1}{3^r} = \frac{1}{3}$$

$$\frac{1}{2} - \frac{1}{2 \times 3^1} = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} \frac{1}{3^r} = \frac{1}{2} - \frac{1}{2 \times 3^k}$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{1}{3^r} = \sum_{r=1}^{k} \frac{1}{3^r} + \frac{1}{3^{k+1}}$$

$$= \frac{1}{2} - \frac{1}{2 \times 3^k} + \frac{1}{3^{k+1}}$$

$$= \frac{1}{2} - \frac{1}{2} \times 3^{-k} + \frac{1}{3} \times 3^{-k}$$

$$= \frac{1}{2} - \frac{1}{6} \times 3^{-k}$$

$$= \frac{1}{2} - \frac{1}{6 \times 3^k}$$

$$= \frac{1}{2} - \frac{1}{2 \times 3 \times 3^k}$$

$$= \frac{1}{2} - \frac{1}{2 \times 3^{k+1}}$$

which is $\frac{1}{2} - \frac{1}{2 \times 3^n}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

9) b)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{r}{2^r} = \frac{1}{2^1} = \frac{1}{2}$$

$$2 - \frac{1+2}{2^1} = 2 - \frac{3}{2} = \frac{1}{2}$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} \frac{r}{2^r} = 2 - \frac{k+2}{2^k}$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{r}{2^r} = \sum_{r=1}^{k} \frac{r}{2^r} + \frac{k+1}{2^{k+1}}$$

$$= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}}$$

$$= 2 - \frac{2(k+2)}{2\times 2^k} + \frac{k+1}{2^{k+1}}$$

$$= 2 + \frac{-2(k+2)+k+1}{2^{k+1}}$$

$$= 2 + \frac{-2k-4+k+1}{2^{k+1}}$$

$$= 2 + \frac{-k-3}{2^{k+1}}$$

$$= 2 - \frac{k+3}{2^{k+1}}$$

which is $2 - \frac{n+2}{2^n}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

9) c)

Prove true for $n = 1$:

$$\sum_{r=1}^{1}(r+1)2^r = 2 \times 2^1 = 4$$

$1 \times 2^{1+1} = 4$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k}(r+1)2^r = k \times 2^{k+1}$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1}(r+1)2^r = \sum_{r=1}^{k}(r+1)2^r + (k+1+1)2^{k+1}$$

$= k \times 2^{k+1} + (k+1+1)2^{k+1}$

$= 2k \times 2^{k+1} + 2 \times 2^{k+1}$

$= 2 \times 2^{k+1}(k+1)$

$= 2^1 \times 2^{k+1}(k+1)$

$= (k+1)2^{k+2}$

which is $n \times 2^{n+1}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

10) a)

Set $n = 1$ so that $2^2 = \frac{1}{6}(a + b + c)$

Set $n = 2$ so that $2^2 + 3^2 = \frac{1}{3}(4a + 2b + c)$

Set $n = 3$ so that $2^2 + 3^2 + 4^2 = \frac{1}{2}(9a + 3b + c)$

Which gives:

$24 = a + b + c$ (1)

$39 = 4a + 2b + c$ (2)

$58 = 9a + 3b + c$ (3)

Solving simultaneously:

(2) – (1) : $15 = 3a + b$

(3) – (1) : $34 = 8a + 2b$

So $a = 2$, $b = 9$

$c = 24 - 2 - 9 = 13$

So that $a = 2$, $b = 9$ and $c = 13$ as req.

10) b)

Prove true for $n = 1$:

$$\sum_{r=1}^{1}(r+1)^2 = 2^2 = 4$$

$$\frac{1}{6} \times 1(2 + 9 + 13) = \frac{1}{6} \times 24 = 4$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k}(r+1)^2 = \frac{1}{6}k(2k^2 + 9k + 13)$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1}(r+1)^2 = \sum_{r=1}^{k}(r+1)^2 + (k+2)^2$$

$$= \frac{1}{6}k(2k^2 + 9k + 13) + (k+2)^2$$

$$= \frac{1}{6}[k(2k^2 + 9k + 13) + 6(k+2)^2]$$

$$= \frac{1}{6}[2k^3 + 9k^2 + 13k + 6k^2 + 24k + 24]$$

$$= \frac{1}{6}[2k^3 + 15k^2 + 37k + 24]$$

$$= \frac{1}{6}(k+1)(2k^2 + 13k + 24)$$

$$= \frac{1}{6}(k+1)(2(k+1)^2 - 4k - 2 + 13k + 24)$$

$$= \frac{1}{6}(k+1)(2(k+1)^2 + 9k + 22)$$

$$= \frac{1}{6}(k+1)(2(k+1)^2 + 9(k+1) + 13)$$

which is $\frac{1}{6}n(2n^2 + 9n + 13)$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

11)

Prove true for $n = 1$:

$$\sum_{r=1}^{1}(1+\mathrm{i})\,\mathrm{i} = -1+\mathrm{i}$$

$$-1+\frac{1}{2}\times 1(1+1)\mathrm{i} = -1+\mathrm{i}$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k}(r+\mathrm{i})\,\mathrm{i} = -k+\frac{1}{2}k(k+1)\mathrm{i}$$

Prove true for $n = k+1$:

$$\sum_{r=1}^{k+1}(r+\mathrm{i})\,\mathrm{i} = \sum_{r=1}^{k}(r+\mathrm{i})\,\mathrm{i} + (k+1+\mathrm{i})\mathrm{i}$$

$$= -k+\tfrac{1}{2}k(k+1)\mathrm{i} + (k+1+\mathrm{i})\mathrm{i}$$

$$= -k+\tfrac{1}{2}k^2\mathrm{i}+\tfrac{1}{2}k\mathrm{i}+k\mathrm{i}+\mathrm{i}-1$$

$$= -(k+1)+\left(\tfrac{1}{2}k^2+\tfrac{3}{2}k+1\right)\mathrm{i}$$

$$= -(k+1)+\tfrac{1}{2}(k^2+3k+2)\mathrm{i}$$

$$= -(k+1)+\tfrac{1}{2}(k+1)(k+2)\mathrm{i}$$

which is $-n+\frac{1}{2}n(n+1)\mathrm{i}$ with $k+1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k+1$, so the statement is true for all integers $n \geq 1$

12)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} r \times r! = 1 \times 1! = 1$$

$$(1+1)! - 1 = 2 - 1 = 1$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} r \times r! = (k+1)! - 1$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} r \times r! = \sum_{r=1}^{k} r \times r! + (k+1) \times (k+1)!$$

$$= (k+1)! - 1 + (k+1) \times (k+1)!$$

$$= [(k+1)! + (k+1) \times (k+1)!] - 1$$

$$= (k+1)! \, [(k+1) + 1] - 1$$

$$= (k+1)! \, (k+2) - 1$$

$$= (k+2)! - 1$$

which is $(n+1)! - 1$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

13)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{r}{(r+1)!} = \frac{1}{2!} = \frac{1}{2}$$

$$1 - \frac{1}{(1+1)!} = 1 - \frac{1}{2} = \frac{1}{2}$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} \frac{r}{(r+1)!} = 1 - \frac{1}{(k+1)!}$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} \frac{r}{(r+1)!} = \sum_{r=1}^{k} \frac{r}{(r+1)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{k+2}{(k+2)(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{k+2}{(k+2)!} + \frac{k+1}{(k+2)!}$$

$$= 1 + \frac{-k-2+k+1}{(k+2)!}$$

$$= 1 - \frac{1}{(k+2)!}$$

which is $1 - \frac{1}{(n+1)!}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

14)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} \frac{1}{(r+a)(r+a+1)} = \frac{1}{(1+a)(2+a)}$$

$$\frac{1}{(1+a)(1+1+a)} = \frac{1}{(1+a)(2+a)}$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} \frac{1}{(r+a)(r+a+1)} = \frac{k}{(1+a)(k+1+a)}$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k} \frac{1}{(r+a)(r+a+1)} = \sum_{r=1}^{k} \frac{1}{(r+a)(r+a+1)} + \frac{1}{(k+1+a)(k+1+a+1)}$$

$$= \frac{k}{(1+a)(k+1+a)} + \frac{1}{(k+1+a)(k+1+a+1)}$$

$$= \frac{k(k+2+a)}{(1+a)(k+1+a)(k+2+a)} + \frac{1+a}{(1+a)(k+1+a)(k+2+a)}$$

$$= \frac{k^2+2k+ak+1+a}{(1+a)(k+1+a)(k+2+a)}$$

$$= \frac{(k+1+a)(k+1)}{(1+a)(k+1+a)(k+2+a)}$$

$$= \frac{(k+1)}{(1+a)(k+2+a)}$$

$$= \frac{k+1}{(1+a)(k+2+a)}$$

which is $\frac{n}{(1+a)(n+1+a)}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

15)

Prove true for $n = 1$:

$$\sum_{r=1}^{1} a^r = a^1 = a$$

$$\frac{a(a^1 - 1)}{a - 1} = \frac{a(a - 1)}{a - 1} = a$$

so true for $n = 1$

Assume true for $n = k$:

$$\sum_{r=1}^{k} a^r = \frac{a(a^k - 1)}{a - 1}$$

Prove true for $n = k + 1$:

$$\sum_{r=1}^{k+1} a^r = \sum_{r=1}^{k+1} a^r + (k+1)^r$$

$$= \frac{a(a^k - 1)}{a - 1} + a^{k+1}$$

$$= \frac{a(a^k - 1)}{a - 1} + \frac{(a-1)a^{k+1}}{a - 1}$$

$$= \frac{a \times a^k - a + a \times a^{k+1} - a^{k+1}}{a - 1}$$

$$= \frac{a^{k+1} - a + a \times a^{k+1} - a^{k+1}}{a - 1}$$

$$= \frac{-a + a \times a^{k+1}}{a - 1}$$

$$= \frac{a(a^{k+1} - 1)}{a - 1}$$

which is $\frac{a(a^n - 1)}{a - 1}$ with $k + 1$ substituted in.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$